

AI-Based PCB Defect Detection System for Manufacturing Quality Control

Problem Statement

PCB manufacturing industries face quality control problems due to defects such as missing components, soldering errors, broken traces, and component misalignment. Manual inspection and traditional Automated Optical Inspection (AOI) systems are often inaccurate, slow, and generate high false-positive rates, leading to increased production costs and reduced product reliability.

Problem Description

Modern Printed Circuit Boards (PCBs) contain hundreds or thousands of tiny electronic components packed into small areas. Detecting manufacturing defects manually is difficult and time-consuming.

Common PCB defects include:

- Missing components
- Misaligned components
- Solder bridges
- Broken copper traces
- Surface scratches

Traditional inspection systems struggle to accurately identify these defects under different lighting conditions and PCB designs. This causes defective products reaching customers, increased rework costs, slower production lines, and higher technician workload.

Proposed Solution

Develop an AI-powered computer vision system that automatically detects PCB defects from captured images in real time. The system captures PCB images using a camera, processes them using AI models, detects and classifies defects automatically, and displays results on a monitoring dashboard.

AI Models Used

1. YOLOv8 – Real-time object detection model used to detect defect locations and classify defects such as missing components, solder bridges, and broken traces.
2. OpenCV – Computer vision library used for image preprocessing, resizing, noise reduction, and edge enhancement.
3. PyTorch – Deep learning framework used to train and optimize the YOLOv8 model.

System Workflow

1. Camera captures PCB image
2. OpenCV preprocesses image
3. YOLOv8 analyzes image
4. Defects are detected and classified
5. Results displayed on dashboard
6. Defective PCB flagged for inspection

Technologies Used

Category	Technology
Programming Language	Python
AI Model	YOLOv8
Computer Vision	OpenCV
Deep Learning	PyTorch
Dashboard	Streamlit or Flask

Expected Outcomes

- Faster PCB inspection
- Higher defect detection accuracy
- Reduced false positives
- Lower manufacturing costs
- Improved product quality

Why This Project Is Achievable

Public PCB defect datasets are available, pretrained YOLOv8 models can be fine-tuned, and the system only requires image-based AI detection. The project can be demonstrated using a webcam or sample PCB images, making it suitable for a university-level implementation.